

Code :

```
def caesar_cipher(text, key):
    result = ""
    for char in text:
        if char.isalpha():
            shift = 65 if char.isupper() else 97
            result += chr((ord(char) - shift + key) % 26 + shift)
        else:
            result += char
    return result

text = str(input("Enter the Plain Text: "))
encrypted = caesar_cipher(text, 3)
decrypted = caesar_cipher(encrypted, -3)
print("Original Text:", text)
print("Encrypted Text:", encrypted)
print("Decrypted Text:", decrypted)
```

Output:

The screenshot displays a code editor with the following content:

EXPLORER

- ...
 - Welcome
 - Caesar_cipher.py X
 - Monoalphabetic_Cipher.py

Caesar_cipher.py > ...

```

1 def caesar_cipher(text, key):
2     result = ""
3     for char in text:
4         if char.isalpha():
5             shift = 65 if char.isupper() else 97
6             result += chr((ord(char) - shift + key) % 26 + shift)
7         else:
8             result += char
9     return result
10
11 text = str(input("Enter the Plain Text: "))
12 encrypted = caesar_cipher(text, 3)
13 decrypted = caesar_cipher(encrypted, -3)
14
15 print("Original Text:", text)
16 print("Encrypted Text:", encrypted)
17 print("Decrypted Text:", decrypted)
18

```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 3

```

PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> & C:/Users/thava/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe C:/Users/thava/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe C:\Users\thava\OneDrive\Desktop\Currently working\SC\Caesar_cipher.py
Enter the Plain Text: THAVAMANI
Original Text: THAVAMANI
Encrypted Text: WKDYDPDQL
Decrypted Text: THAVAMANI
PS C:\Users\thava\OneDrive\Desktop\Currently working\SC>

```